

Chapter 1

SET THEORY

[Part 1: Set & Subset]

Introduction

Why are we studying sets

- The concept of set is basic to all of mathematics and mathematical applications.
- Serves as a basis of description of higher concept and mathematical reasoning
- Set is fundamental in many areas of Computer Science.

Set

- A set is a **well-defined collection of distinct objects.**
- These objects are called **members** or **elements** of the set.
- Well-defined means that we can tell for certain whether an object is a member of the collection or not.
- If a set is finite and not too large, we can describe it by listing the elements in it.

Example

- A is a set of all positive integers less than 10,
 $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- B is a set of first 5 positive odd integers,
 $B = \{1, 3, 5, 7, 9\}$
- C is a set of vowels, $C = \{a, e, i, o, u\}$

Defining Sets

This can be done by:

- Listing ALL elements of the set within $\{ \}$ braces.)
- Listing enough elements to show the pattern then an ellipsis. $(\dots)$
- Use set builder notation to define “rules” for determining membership in the set

Example

1. Listing ALL elements. $A = \{1, 2, 3, 4\}$ explicitly
2. Demonstrating a pattern. $\mathbb{N} = \{1, 2, 3, \dots\}$ implicitly
3. Using set builder notation. $P = \{x \mid x \in \mathbb{R} \text{ and } x \notin \mathbb{C}\}$ implicitly

Sets

A set is determined by its elements and not by any particular order in which the element might be listed.

Example, $A = \{1, 2, 3, 4\}$,

A might just as well be specified as
 $\{2, 3, 4, 1\}$ or $\{4, 1, 3, 2\}$

Sets

The elements making up a set are assumed to be **distinct**, we may have duplicates in our list, only one occurrence of each element is in the set.

Example

$$\{a, b, c, a, c\} \longrightarrow \{a, b, c\}$$

$$\{1, 3, 3, 5, 1\} \longrightarrow \{1, 3, 5\}$$

Sets

- Use uppercase letters $A, B, C \dots$ to denote sets, lowercase denote the elements of set.
- The symbol $\in$ stands for 'belongs to'
- The symbol $\notin$ stands for 'does not belong to'

Example

$$X = \{ a, b, c, d, e \}, \quad b \in X \text{ and } m \notin X$$

$$A = \{ \{1\}, \{2\}, 3, 4 \}, \quad \{2\} \in A \text{ and } 1 \notin A$$

- If a (set is a large finite set or an infinite set,) we can describe it by **listing a property necessary for memberships**

- Let S be a set, the notation,

$$A = \{x \mid x \in S, P(x)\} \text{ or } A = \{x \in S \mid P(x)\}$$

means that A is the set of all elements x of S such that x satisfies the property P .

Example

- Let $A = \{1, 2, 3, 4, 5, 6\}$, we can also write A as,

$A = \{x \mid x \in \mathbb{Z}, 0 < x < 7\}$ if $\mathbb{Z}$ denotes the set of integers.

- Let $B = \{x \mid x \in \mathbb{Z}, x > 0\}$, $B = \{1, 2, 3, 4, \dots\}$

Example

The set of natural numbers: $\mathbb{N} = \{1, 2, 3, \dots\}$

The set of integers: $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

The set of positive integers: $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$

The set of Rational Numbers (fractions): $\frac{1}{2}, \frac{2}{3}, \frac{5}{7}, \text{etc} \in \mathbb{Q}$

More formally: $\mathbb{Q} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{R}, b \neq 0 \right\}$

The set of Irrational Numbers: $\sqrt{2}, \pi$, or e are irrational

The Real numbers $= \mathbb{R}$ = the union of the rational numbers
with the irrational numbers

Some Symbols Used With Set Builder Notation

The standard form of notation for this is called "set builder notation".

For instance, $\{x \mid x \text{ is an odd positive integer}\}$ represents the set $\{1, 3, 5, 7, 9, \dots\}$

$\{x \mid x \text{ is an odd positive integer}\}$ is read as

"the set consisting of all x such that x is an odd positive integer".

The vertical bar, " $\mid$ ", stands for "such that"

Other "short-hand" notation used in working with sets

" $\forall$ " stands for "for every"

" $\cup$ " stands for "union"

" $\subseteq$ " stands for "is a subset of"

" $\subsetneq$ " stands for "is a not a (proper) subset of"

" $\in$ " stands for "is an element of"

" $\times$ " stands for "cartesian cross product"

" $\exists$ " stands for "there exists"

" $\cap$ " stands for "intersection"

" $\subset$ " stands for "is a (proper) subset of"

" $\emptyset$ " stands for the "empty set"

" $\notin$ " stands for "is not an element of"

" $=$ " stands for "is equal to"

Subset

If every element of A is an element of B , we say that A is a subset of B and write $A \subseteq B$.

$A=B$, if $A \subseteq B$ and $B \subseteq A$

The **empty set** ($\emptyset$) is a subset of every set.

Example $A=\{1, 2, 3\}$

Subset of A ,

$\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$

Note: A is a subset of A

Proper Subset

If $A \subseteq B$ and B contains an element that is not in A , then we say “ A is a **proper subset** of B ”: $A \subset B$ or $B \supset A$.

Formally: $A \subseteq B$ means $\forall x [x \in A \rightarrow x \in B]$.

For all sets: $A \subseteq A$.

Note: If A is a subset of B and A does not equal B , we say that A is a proper subset of B ($A \subseteq B$ and $A \neq B$ ($B \not\subseteq A$))

Example

- Let, $A = \{1, 2, 3\}$

Proper subset of A ,

$\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}$

- Let, $B = \{1, 2, 3, 4, 5, 6\}$

A is proper subset of B .

Example

$A = \{a, b, c, d, e, f, g, h\}$

$B = \{b, d, e\}$

$C = \{a, b, c, d, e\}$

$D = \{r, s, d, e\}$

Proper subset of A ??

Empty Sets

The **empty set** $\emptyset$ or $\{\}$ **but not** $\{\emptyset\}$
is the set without elements.

Note:

- Empty set has no elements
- Empty set is a subset of any set
- There is exactly one empty set
- Properties of empty set:

$$A \cup \emptyset = A, A \cap \emptyset = \emptyset$$

$$A \cap A' = \emptyset, A \cup A' = U$$

$$U' = \emptyset, \emptyset' = U$$

Example

$$\emptyset = \{x \mid x \text{ is a real number and } x^2 = -3 \}$$

$$\emptyset = \{x \mid x \text{ is positive integer and } x^3 < 0\}$$

Equal Sets

The sets A and B are **equal** ($A=B$) if and only if each element of A is an element of B and vice versa.

Formally: $A=B$ means $\forall x [x \in A \leftrightarrow x \in B]$.

Example

$$A=\{a, b, c\}, B=\{b, c, a\}, \quad A=B$$

$$C=\{1, 2, 3, 4\}$$

$$D=\{x \mid x \text{ is a positive integer and } 2x < 10\},$$

$$C=D$$

Equivalent Sets

Two sets, A and B, are **equivalent** if there exists a **one-to-one correspondence** between them.

When we say sets “have the same size”, we mean that they are **equivalent**.

Example

Set A = {A, B, C, D, E} and **Set B** = {1, 2, 3, 4, 5}

Note:

- An equivalent set is simply a set with an **equal number of elements**.
- The sets do not have to have the same exact elements, just the same number of elements.

Finite Sets

A set A is **finite**

if it is empty

or

if there is a natural number n
such that set A is equivalent to

$\{1, 2, 3, \dots n\}$.

Example

$$A = \{1, 2, 3, 4\}$$

$$B = \{x \mid x \text{ is an integer, } 1 \leq x \leq 4\}$$

Note:

There exists a nonnegative integer n such that A has n elements (A is called a finite set with n elements)

Infinite Sets

- An infinite set is a set whose **elements can not be counted.**
- An infinite set is one that has **no last element**

Are all infinite sets equivalent?

An infinite set is a set that can be placed into a **one-to-one correspondence** with a proper subset of itself.

Example

Infinite sets

$$Z = \{x \mid x \text{ is an integer}\}$$

$$\text{or } Z = \{..., -3, -2, -1, 0, 1, 2, 3, ...\}$$

$$S = \{x \mid x \text{ is a real number and } 1 \leq x \leq 4\}$$

$$D = \{x \mid x \text{ is an integer, } x > 0\}$$

Finite Sets

$$C = \{5, 6, 7, 8, 9, 10\}$$

$$B = \{x \mid x \text{ is an integer, } 10 < x < 20\}$$

Universal Set

- Sometimes we are dealing with sets all of which are subsets of a set U .
- This set U is called a universal set or a universe.
- The set U must be explicitly given or inferred from the context

Universal Set

Typically we consider a set A
a part of a **universal set $\mathcal{U}$** ,
which consists of all possible elements.

To be entirely correct we should say

$$\forall x \in \mathcal{U} [x \in A \leftrightarrow x \in B]$$

instead of

$$\forall x [x \in A \leftrightarrow x \in B] \text{ for } A=B.$$

Note that $\{ x \mid 0 < x < 5 \}$ is can be ambiguous.

Compare $\{ x \mid 0 < x < 5, x \in \mathbb{N} \}$ with $\{ x \mid 0 < x < 5, x \in \mathbb{Q} \}$

Example

- The sets $A=\{1,2,3\}$, $B=\{2,4,6,8\}$ and $C=\{5,7\}$
- One may choose $U=\{1,2,3,4,5,6,7,8\}$ as a universal set.
- Any superset of U can also be considered a universal set for these sets A , B , and C .

For example, $U=\{x \mid x \text{ is a positive integer}\}$

Cardinality of Set

- Let S be a finite set with n distinct elements, where $n \geq 0$.
- Then we write $|S|=n$ and say that the **cardinality** (or **the number of elements**) of S is n .

Example

$$A = \{1, 2, 3\}, \quad |A|=3$$

$$B = \{a, b, c, d, e, f, g\}, \quad |B|=7$$

Power Set

- The set of all subsets of a set A , denoted $P(A)$, is called the **power set of A** .

$$P(A) = \{X \mid X \subseteq A\}$$

$$\text{If } |A|=n, \text{ then } |P(A)| = 2^n$$

Example $A = \{1, 2, 3\}$

The power set of A ,

$$P(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

Notice that $|A| = 3$, and $|P(A)| = 2^3 = 8$

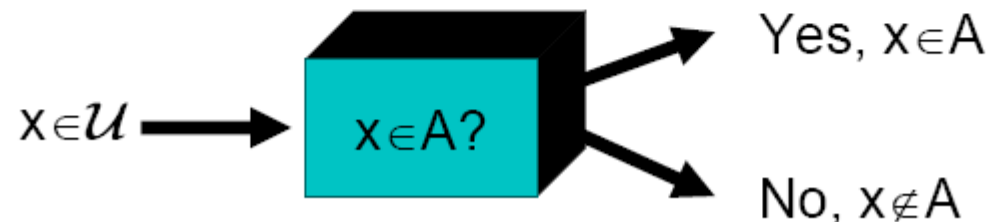
How to Think of Sets

The elements of a set do not have an ordering,
hence $\{a,b,c\} = \{b,c,a\}$

The elements of a set do not have multitudes,
hence $\{a,a,a\} = \{a,a\} = \{a\}$

All that matters is: “Is x an element of A or not?”

The size of A is thus the number of *different* elements



Thank You